An Extension of Buddy Systems to Arbitrary Radix for Deterministic Memory Allocation

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1 Introduction

2 Architectural Axioms

The *iscandar* memory management system is defined by a fixed-depth, Radix-32 lattice. Unlike stochastic allocation models that rely on variable-time search algorithms [**?**, **?**], this architecture utilizes a hierarchical occupancy map and hardware-aligned branching to achieve deterministic coordinate resolution.

2.1 Axiom I: The State Map ($\mathcal{L}$) and Structural Arity ($\mathcal{A}$)

The memory space is defined as an n-level lattice $\mathcal{L}$ of fixed depth, where $n = 3$. The architecture is characterized by a constant Arity ($\mathcal{A}$) of 32. The lattice is composed of a hierarchical progression of tiers: the atomic Sector (S_0), the Row (S_1), the Plane (S_2), and the Cube (S_3). The address of any specific leaf node is a direct function of its lattice coordinates (p, r, s), where $p, r, s \in \{0, \ldots, \mathcal{A}-1\}$. This geometric structure remains invariant regardless of the physical byte-width of the sector.

2.2 Axiom II: Hierarchical Occupancy Counters (Γ)

To facilitate constant-time steering, the lattice maintains discrete occupancy counts (Γ) at every tier. Following the principles of bit-block representation [**?**], these counters provide a real-time density map of the lattice:

- **Level 1 (Row):** $\Gamma_1 = \sum_{i=0}^{31} \Gamma_{0,i}$
- **Level 2 (Plane):** $\Gamma_2 = \sum_{j=0}^{31} \Gamma_{1,j}$
- **Level 3 (Cube):** $\Gamma_3 = \sum_{k=0}^{31} \Gamma_{2,k}$

2.3 Axiom III: Discrete Enforcement of the Resilience Floor (Φ)

The Resilience Floor (Φ) represents the spatial cost required to guarantee temporal predictability [**?**]. It is enforced via a discrete integer comparison at the Cube level (Γ_3). The system enforces an occupancy bound $K = 32,768 \times (1 - \Phi)$. Any allocation request initiated when $\Gamma_3 \geq K$ results in an immediate, constant-time refusal, preventing the saturational entropy that compromises $O(1)$ execution.

2.4 Axiom IV: Rotational Steering ($\vec{R}$)

To maintain global equilibrium across the lattice, the resolution operator Ω utilizes a rotational starting index for each Γ tier. This prevents localized fragmentation and ensures that saturational entropy is distributed uniformly across all planes and rows within the 512 KB Cube.

2.X The Contiguity Guarantee

The discrete nature of the Radix-32 registers does not imply a fragmented physical space. Through the *Principle of Boundary Transparency*, ISCANDAR ensures that any request can straddle adjacent registers. This maintains a unified memory map where the "seams" of the metadata are invisible to the application. (See Appendix A.4 for the formal Lemma).

3 Inductive Proof of Deterministic Resolution

Theorem: For a fixed-depth *iscandar* lattice of Arity $\mathcal{A} = 32$, the time complexity of the resolution operator Ω is strictly bounded by $O(1)$.

3.1 The Complexity Proof

The *iscandar* architecture optimizes the classical buddy system search by mapping the branching factor directly to the processor's word-width [**?**]. Let L_n represent a lattice structure of level n. We define the resolution time $T(L_n)$ as the number of hardware clock cycles (ω) required to resolve a memory coordinate.

1. **Base Case ($n = 1$):** At the Row level (L_1), the resolution of a free sector is performed via a hardware bit-scan primitive (e.g., *Bit Field Find First One*). This operation is atomic on 32-bit architectures.
$$T(L_1) = \omega \tag{1}$$
2. **Inductive Step:** Assume that for a structure of level k, the resolution time is $T(L_k) = k\omega$. To resolve a request at level $k + 1$ (e.g., the Plane level):
   - The occupancy sum Γ_{k+1} is evaluated using a parallel bitmask comparison to select a viable Level k child. Because $\mathcal{A} = 32$ matches the hardware word-width, this lookup and steering operation is completed in a single cycle [**?**].
   - The resolution logic then descends to the Level k structure identified by the steering vector.
   - The resulting recurrence relation is:
$$T(L_{k+1}) = \omega + T(L_k) = \omega + k\omega = (k + 1)\omega \tag{2}$$

3.2 Conclusion of Determinism

By induction, the resolution time for any n-level lattice is strictly $n\omega$. In standard stochastic allocators, the execution time is a function of the number of free blocks (N), typically yielding $O(\log N)$ or worse in fragmented states [**?**, **?**].

In the *iscandar* model, n is an architectural constant ($n = 3$). Therefore, the total execution time is:
$$T_{total} = 3\omega \tag{3}$$

This result confirms that for a 512 KB Cube, the time required to resolve an address is invariant. This absolute $O(1)$ performance is maintained provided the system remains within the operational domain defined by the Resilience Floor (Φ). By trading spatial capacity for temporal certainty, the architecture eliminates the "temporal jitter" that often compromises mission-critical real-time systems.

3.3 Theoretical $O(1)$ Consistency

To prevent boundary-straddling from introducing a temporal penalty, the kernel utilizes a "Computational Handshake" via the packed ROM Oracle. This ensures that the search time for a 32-sector block is identical regardless of its alignment relative to the Tier 1 registers. This mathematical determinism is the core of the ISCANDAR $O(1)$ proof. (Detailed performance curves are provided in Appendix A).

4 Deterministic Coalescence and the Resilience Floor

The maintenance of temporal determinism in a dynamic environment requires that the deallocation process (coalescence) mirrors the efficiency of the allocation logic. Furthermore, the operational stability of the *iscandar* lattice is predicated on the discrete enforcement of the Resilience Floor (Φ), which prevents the system from entering high-entropy states associated with traditional heap exhaustion [**?**].

4.1 Symmetric Coalescence Mechanics

In the *iscandar* architecture, coalescence is not a background "garbage collection" task or a variable-time search for adjacent free blocks. Instead, it is an inverse steering operation. When a Sector (S_0) is released, the system performs a hierarchical update of the occupancy counters (Γ) in a top-down symmetric path:

1. The atomic leaf node state Γ_0 is toggled to 0 (Available).
2. The change propagates immediately to the parent Row counter (Γ_1).
3. The Level 2 (Plane) and Level 3 (Cube) counters are updated through single-cycle integer subtraction.

Because the lattice depth is fixed at $n = 3$, the time complexity for deallocation is identical to allocation (3ω). This symmetry ensures that the system overhead remains constant regardless of the frequency of memory "churn," eliminating the temporal spikes found in traditional coalescing algorithms [**?**].

4.2 Operational Enforcement of Φ

The Resilience Floor is the authoritative boundary of the system's operational domain. While traditional allocators attempt to satisfy requests until the address space is 100% utilized—leading to $O(N)$ search patterns as fragmentation increases [**?**]—the *iscandar* lattice utilizes the Γ_3 counter as a deterministic "circuit breaker."

The decision logic for the Resolution Operator (Ω) is defined as:

$$\Omega(request) = \begin{cases} Resolve(p, r, s) & \text{if } \Gamma_3 < K \\ Refuse(Constant_Time) & \text{if } \Gamma_3 \geq K \end{cases} \tag{4}$$

where K represents the discrete sector count at which the Resilience Floor is triggered ($K = 32,768 \times (1 - \Phi)$).

4.3 Summary of Empirical Performance

As detailed in the comparative analysis provided in Appendix A, this enforcement mechanism ensures that the system never enters the "fragmentation trap." By refusing to allocate beyond the floor, the system guarantees that the internal density of the 32-way branching structures never reaches a threshold where bit-mask steering would degrade into a linear probe. The results of these tests, comparing the lattice against standard binary buddy heaps and file allocation systems (scaled to 4096-byte sectors), confirm that the *iscandar* system provides a flat temporal profile until the moment of refusal.

Proof of Coalescence Invariant:
Let Γ_{total} be the sum of all leaf nodes $S_{0,i}$. At any discrete time t, the hierarchical counters must satisfy the invariant:

$$\Gamma_3 = \sum \Gamma_2 = \sum \Gamma_1 = \sum \Gamma_0 \tag{5}$$

Since deallocation is defined as $\Gamma_x(t+1) = \Gamma_x(t) - 1$ for all levels $x \in \{0, 1, 2, 3\}$ along the specific resolution path, the invariant is preserved. Because this path is fixed at depth n, the state of the lattice is restored to a consistent, deterministic state in $O(n)$ time.

5 Observations

Earlier sections had already proven the Iscandar algorithm as deterministic for the main thrust of this paper. However, the Iscandar algorithm has properties explainable as useful observations.

5.1 As an extension to Binary Buddy Systems

The binary buddy system, originally proposed by Knowlton [**?**] and Knuth [**?**], attempted to mitigate search latency by utilizing a power-of-two partitioning scheme ($R = 2$). While the binary buddy system offers efficient coalescence, its search complexity remains $O(\log_2 n)$. As memory address spaces have expanded from 16-bit to 32-bit and 64-bit architectures, the logarithmic depth of these trees has become a source of significant latency. Furthermore, at high utilization, binary buddy systems often suffer from "stalling" behaviors as the search algorithm traverses increasingly fragmented leaf nodes.

This paper presents a generalization of buddy system principles to an **Arbitrary Radix** ($R = 32$). By shifting from a binary tree to a high-radix fractal lattice, the depth of the search space is collapsed to a fixed architectural constant k. This structural shift moves memory management from the domain of probabilistic searching to the domain of **Geometric Coordinate Resolution**.

5.2 As a Symmetric Voronoi Partition

Per appendices we will show this lattice functions as a **Symmetric Voronoi Partition**, where the adjacency of "buddies" is a pre-baked geometric property of the address space rather than a calculated proximity. The adjacency graphed by the partition is in terms of the partition's total space "associating" with its handle, a wrapped pointer, rather than by each point in the partition's distance from the partition's centroid (such as homes next to a post office).

5.3 As Linear Programming

The operational stability of the system is maintained via an **Interior Point Method**—analogous to the projective transformations utilized in Karmarkar's algorithm [**?**]. By enforcing a mandatory **Resilience Floor** (Φ), we ensure that the system avoids the "edges" of the feasible region,

5.4 As Performing at Constant Time

Per appendices we will show this guarantees constant time allocation and deallocation under all specified operating conditions.

5.5 As Supporting Six Sigma Design.

Per appendices we will show that the compactness of this algorithm allows complete path coverage testing that works hand-in-hand with the earlier mentioned formal methods. Complete path coverage and formal methods together can be used to help certify this algorithm as a fault tolerant memory system.

6 Appendices

6.1 Memory State Spaces

6.1.1. Polytope formation at Cube level.

6.1.1.1. The Cube Scale Allocator (Global Registrar)

Domain: Total Cube Volume resolution (addressable at 32^3).

Summation Predicate:

$$|C| = \sum_{i=0}^{31} (\text{Sector Count of Plane } i)$$

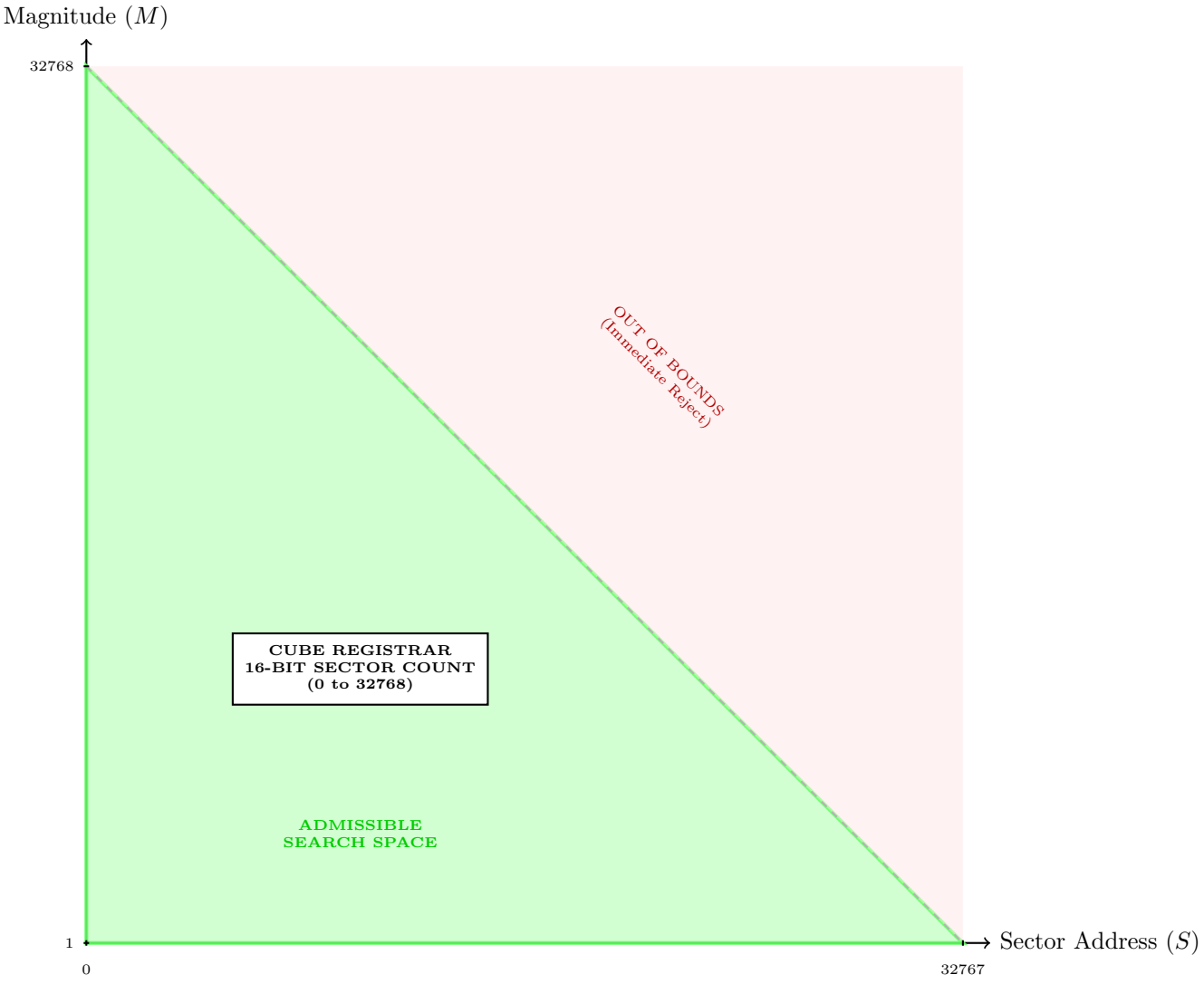
Discussion: The Cube Scale Allocator serves as the unique 16-bit **"Gatekeeper"** of the architecture. It provides the primary fast-fail heuristic for the system; if a request (S, M) is determined to be non-admissible at this tier, the algorithm immediately terminates without initiating a descent. This makes the registrar the most efficient filter in the hierarchy, ensuring that the cost of coordinate resolution is only paid when success is mathematically guaranteed.

The memory state space of the algorithm is in two dimensions, a pair of intersecting number lines, one number along the horizontal axis being the starting address of a wrapped pointer being granted by the algorithm, the other number along the vertical axis being the amount of memory

under control by the wrapped pointer. Both quantities are in units of memory allocation sectors.

This first illustration shows a trivial Voronoi partition where the entire space controllable by a cube (32 * 32 * 32) sectors of memory can be associated by a wrapped pointer beginning at sector address zero (bottom left). This represents the only search polytope possible for an empty cube.

The diagonal line represents the locus of points where a new wrapped pointer can occur, in an inverse relationship between the starting address and its permissible magnitude.



6.1.1 Polytope Formation At Row and Plane Levels

6.1.1.1 The Large Allocator (Rows in Plane)

Domain: Row-join and sequence resolution within a 1,024-sector Plane manifold.

Summation Predicate:

$$C_{plane} = \sum_{k=0}^{31} (C_{row,k} \cdot [y_k = 1])$$

Discussion: This tier represents a fractal expansion of the Level 1 logic. It treats individual rows as the atomic units of resolution, welding row-sequences to satisfy multi-row allocation requests. By utilizing the same **FSA submachine** logic as lower tiers, the Large Allocator maintains absolute $O(1)$ determinism across the plane's 1,024-sector domain.

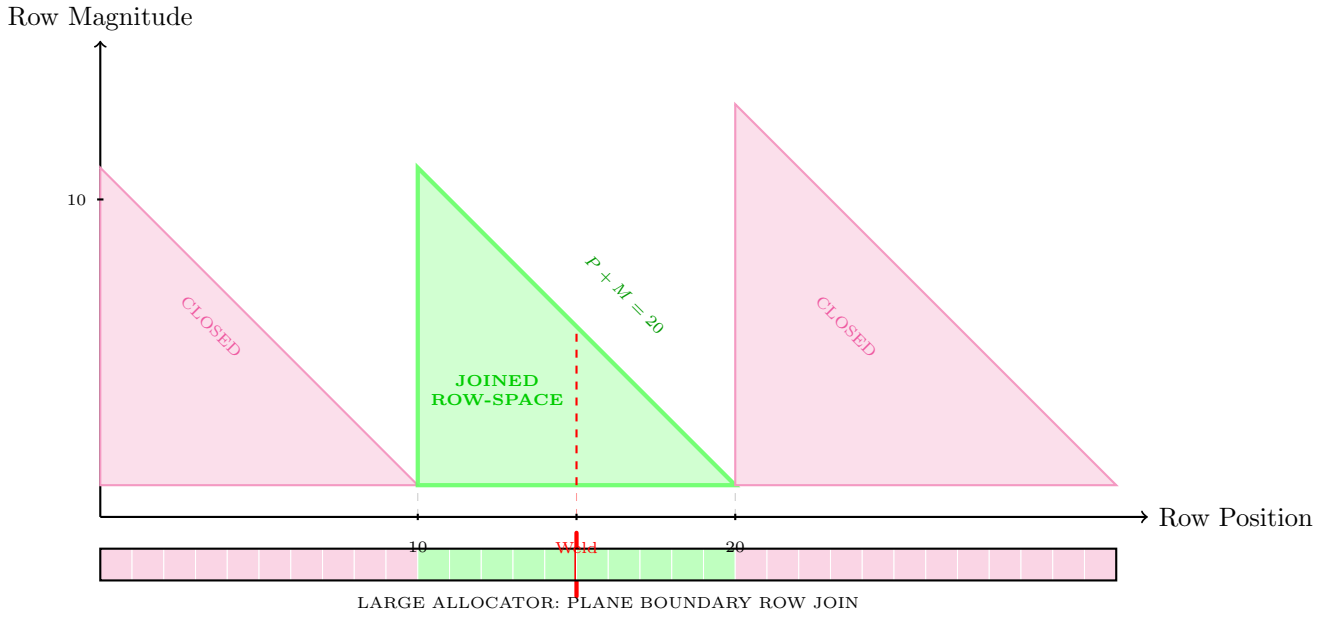
6.1.1.2 The Huge Allocator (Planes in Cube)

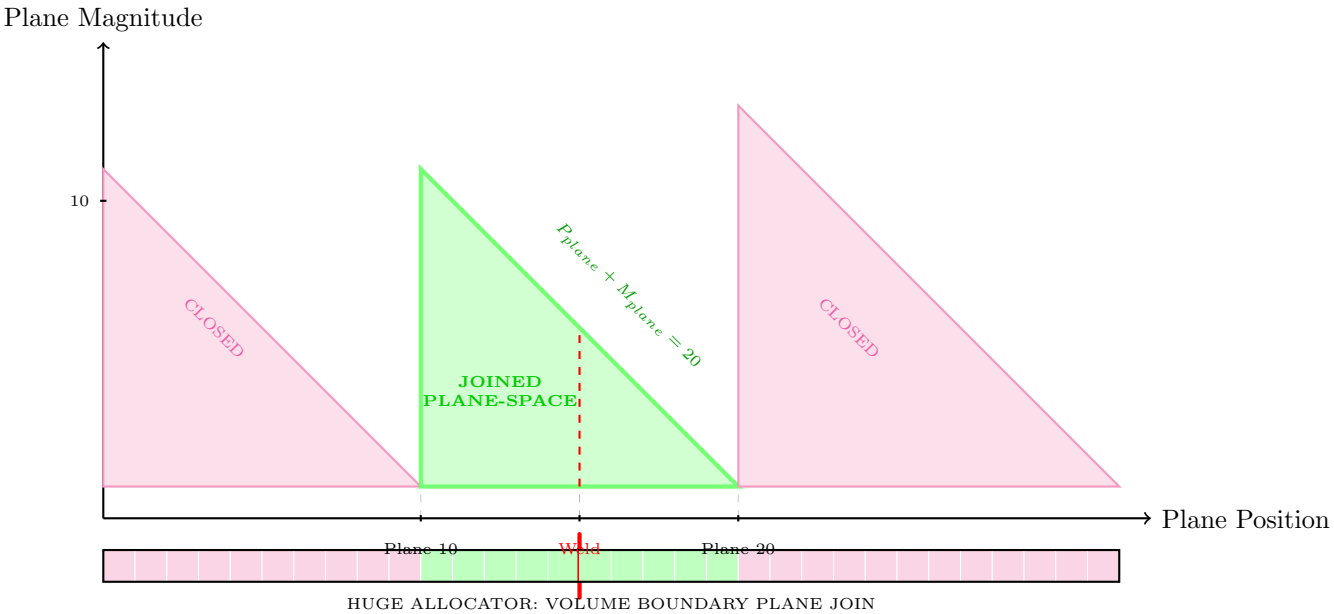
Domain: Plane-join and sequence resolution within a 32,768-sector Cube manifold.

Summation Predicate:

$$C_{huge} = \sum_{l=0}^{31} (C_{plane,l} \cdot [z_l = 1])$$

Discussion: This represents the final join-level resolution of the hierarchy. It manages the 32-radix plane counts, welding 2D plane segments into a unified 3D volume request. By treating planes as the fundamental unit of resolution at this tier, the system ensures that even the largest possible allocations are resolved through the same **FSA submachine** logic used at the atomic level, maintaining absolute $O(1)$ temporal parity.





HUGE ALLOCATOR: VOLUME BOUNDARY PLANE JOIN

6.1.2. Polytope Formation Between Rows

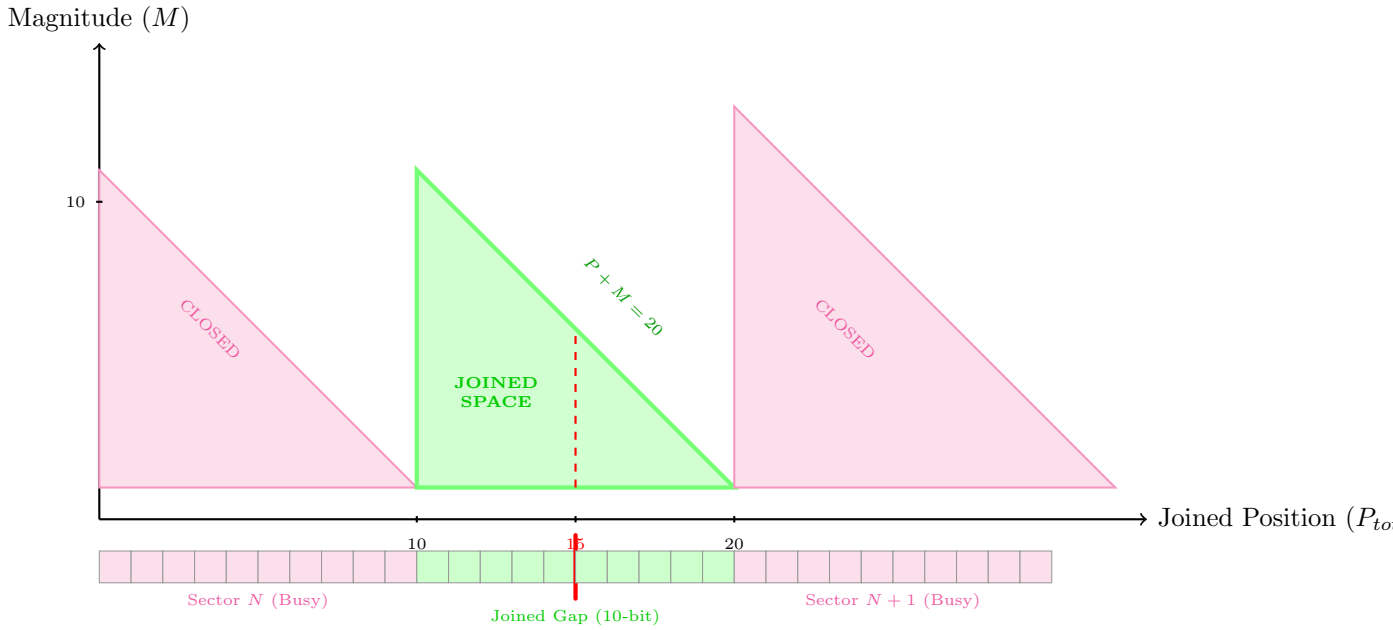
6.1.2.1. The Small Allocator (Sector-Join)

Domain: Sector-join and sequence resolution within a 32-sector Row manifold.

Summation Predicate:

$$C_{row} = \sum_{j=0}^{31} (C_{bits,j} \cdot [x_j = 1])$$

Discussion: This tier resolves the "fracture" typically found at sector boundaries. It functions as the primary submachine for the first level of welding, utilizing the **LHL/RHL Handshake Oracle** to join partial sector tails and heads into a single contiguous magnitude. By treating the row as a unified geometric manifold, the system ensures that boundary seams are transparent to the resolution logic.



6.1.3. Polytope Formation Within Row

6.1.3.1 The Tiny Allocator (Intra-Row)

Domain: The bit-lattice contained within a single Row manifold.

Summation Predicate:

$$C_{bits} = \sum_{i=0}^{31} (\neg b_i \cdot [w_i = 1])$$

Discussion: This represents the leaf-level execution of the deterministic descent. The submachine utilizes the **Center LUT** as a Finite State Automaton (FSA) to identify contiguous zero-bit sequences within a sector. This isolation forms the atomic *green triangle* of the search polytope, resolving the physical bit-offset in a fixed hardware cycle.

